

Project Report: Power Electronics

B1 - Group: 6

Problem Statement: SEPIC Converter

Roll Numbers: 240002014, 240002021, 240002029, 240002041

1 Project Requirements

1.1 Specifications

- **Input Voltage:** 15 V
- **Output Voltage:** 10 – 20 V
- **Switching Frequency:** 5 kHz
- **Output Current:** 1 A

1.2 Objectives

- Observe and document the waveforms of the switch current and diode voltage in Continuous Conduction Mode (CCM).
- Increase the load resistance to successfully demonstrate Discontinuous Conduction Mode (DCM).

2 Hardware Setup

The hardware implementation of the SEPIC converter involves the control circuitry and the primary power stage. Placing the circuits side-by-side reveals the complete hardware integration.

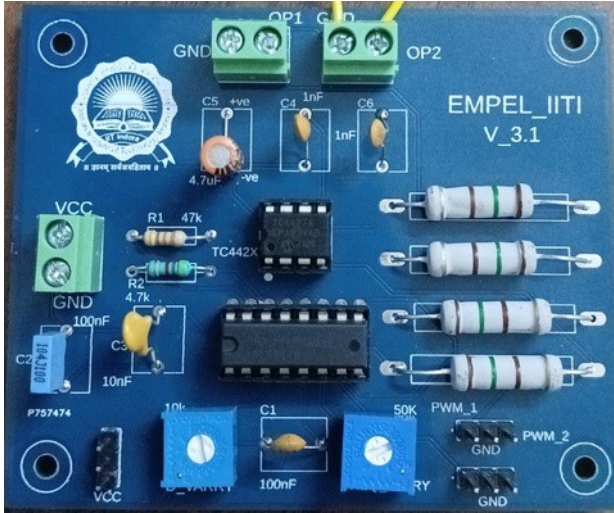


Figure 1: Gate Driver Circuit

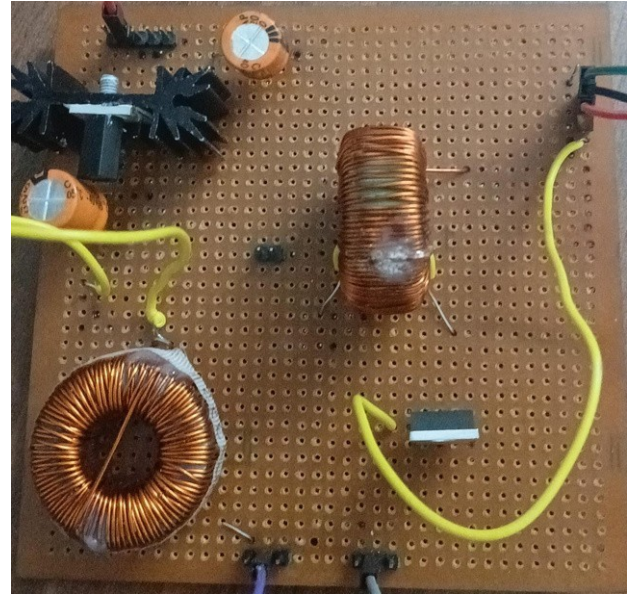


Figure 2: SEPIC Converter soldered on Perf board

3 Output Voltage Verification

The output characteristics were verified under different operating conditions to confirm both buck and boost capabilities.



Figure 3: Demonstration of Output Current at 1 A. The red waveform indicates the output current measurement.

3.1 Operating Modes

The converter was tested in both Buck and Boost modes at a 5 kHz switching frequency with a 15 V input.



Figure 4: Buck Mode ($D < 0.5$): 40% Duty Cycle
Output ≈ 8.13 V



Figure 5: Boost Mode ($D > 0.5$): 54% Duty Cycle
Output ≈ 16.78 V

4 Waveforms in CCM Mode

The following waveforms were captured while the converter was operating in Continuous Conduction Mode (CCM).



Figure 6: Switch Current Waveform in CCM. The cyan waveform represents the gate pulse, and the red waveform represents the switch current.

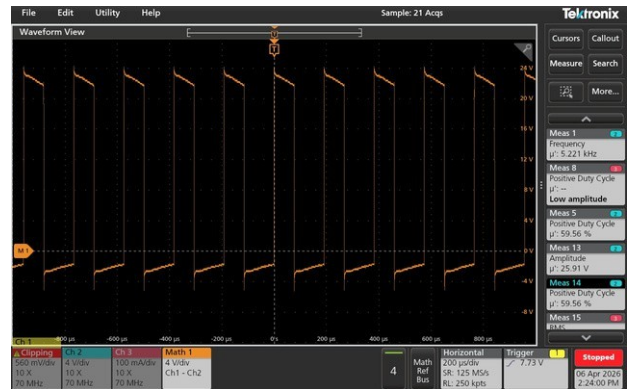


Figure 7: Diode Voltage Waveform in CCM, represented by the calculated orange math waveform.

5 Depiction of Inductor Currents

By adjusting the load resistance, the transition between Continuous Conduction Mode and Discontinuous Conduction Mode was observed in the inductor current.

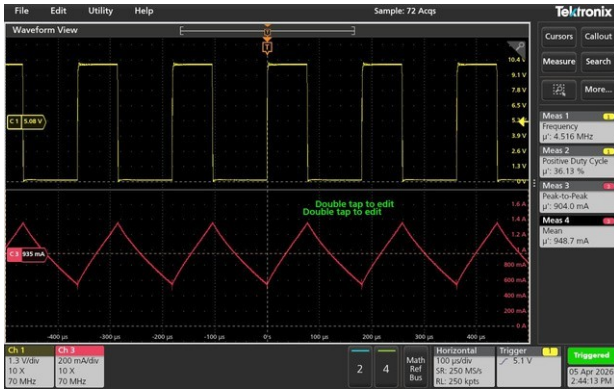


Figure 8: Inductor Current Operating in CCM. The yellow waveform represents the gate pulse, and the red waveform represents the inductor current.



Figure 9: Inductor Current Operating in DCM. The yellow waveform represents the gate pulse, and the cyan waveform represents the inductor current.